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OP.2.GE Project
NAME: ABO WILLIAM JUNIOR

SUBMISSION OF 2nd SEMESTER OF DRONE PROJECT DOMAIN EEE

INTRODUCTION

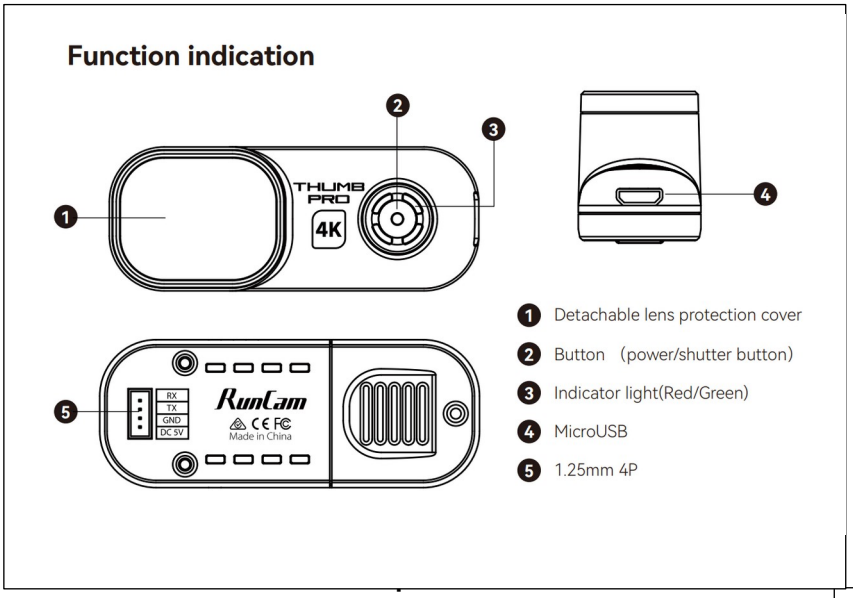
At the end of the 1st semester, The EEE domain allows us to analyze the electronic system of the drone point by point in order to better understand how it works. But still going to review them back in the final power point respectively.

But for this 2nd semester, the work was carried out as part of the drone project in the field of electrical engineering (EEE) concerning the objectives of semester 2. The main deliverables include the addition of a shooting camera on the drone, the transmission and means of viewing the images, the procedure for calibrating sensors and ESCs, as well as the refinement of the calculation of battery life.

This report covers phases 1 and 2 of the Electrical Engineering field, detailing the steps of implementing the camera, the calibration of the sensors and the calculation of the autonomy of the drone in flight

PHASE 1: Characteristics and implementation of the camera on the drone

Before the characteristics we are first of all going to talk of the description of the camera;







- Resolution: 4K@30fps, 2.7K@60fps, 1440P@60fps, 1080P@120fps, 1080P@60fps
- **Sensor**: IMX577 (12MP)
- **F.O.V**: 155°
- **Video Format**: MP4
- **Gyro**: Built-In
- Support Micro SD Card capacity:256G Max



- USB Power Input voltage: DC 5V
- **Remote control**: UART
- **Power Consumption**: 500mA at 5V Max
- **Communication Interface**: Micro USB
- **Dimensions**: 54mm x 25.5mm x 21mm
- **Video capture resolution** : 4K
- **Net Weight**: 16 grams

2-IMPLEMENTATION OF THE CAMERA ON THE DRONE

In this phase, our group focused on adding a camera to the drone and its integration.

The steps taken included the study of the fixation of the camera to the drone, the verification of the possibility of shooting and the quality of the images, as well as the test of image transmission. But before studying the fixation of the camera and the transmission of images, we first made a choice of camera.

-FIXATION ON THE CAMERA ON THE DRONE

Then we studied the implementation of the camera on the drone. We were expected to be able to move our drone with a photographic camera, which allows the drone to take photos and videos of very good resolution during its flight. To achieve this, we first found the ideal camera and know how to attach/equipped it to the drone. To get there, we chose to use a GIMBAL (CAMERA SUPPORT or NACELLE (in French)).

➢ What is a GIMBAL(NACELLE)?

The nacelle, also called a drone stabilizer (camera support) is a mechanical device mounted on the drone that allows the camera to be stabilized and controlled

➢ How does GIMBAL work?

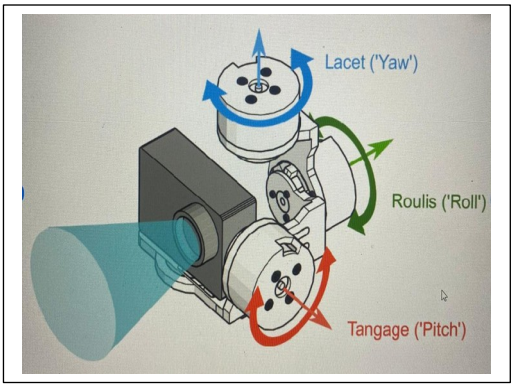
The nacelle usually uses motors and sensors to compensate for undesirable movements such as vibrations, tremors, and inclinations of the drone. The engines are reacting to the signals sent by the sensors and adjusts the position or inclination of the camera to keep it stable.

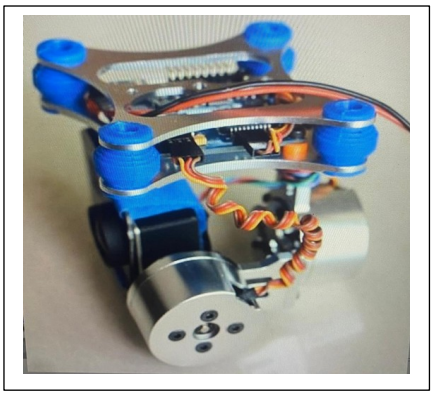
TYPES OF GIMBALS(NACELLE)

We can find a wide variety of nacelles, but here we are going to talk about the following nacelle

- Nacelle with two axes: it is a nacelle designed to provide stability to the camera along two axes, making it possible to maintain 1 straight horizon and attenuate unwanted movements along these axes.

- Nacelle with three axes: it's is a nacelle designed to provide stability to the camera according to three axes, attenuating unwanted movements along these axes while maintaining a straight horizon





- Motorized nacelle: these are nacelles equipped with a motor that allows a controlled movement of the camera. They can be controlled remotely to adjust the angle of the camera during flight, offering great flexibility for image and video recording.



TABLE OF DECISION OF CRITERIA

The following table is a comparative table between different nacelles above to choose the most suitable for our drone.

	Motorized nacelle	Nacelle with 2-axis	Nacelle with 3-axis
Consumption(v)	10 – 25	7.5 – 11.2	12 – 30
Weight(g)	160 – 500	160 – 500	210 – 960
Stabilization	Good	Good	High
Flexibility	Have compatible software	Good flexibility	Very accessible flexibility
Control	Manual	Automatic/Manual	Automatic/Manual

Cost	115 -400$	70-100$	95-500$

CHOICE OF THE NACELLE

In conclusion, we've chosen the 2-axis nacelle.

REASON FOR THE CHOICE

We chose the 2-axis nacelle due to the fact that it offers good stability and flexibility during flight for quality aerial shots, it has an average weight compatible with small and medium-sized drones and an affordable cost. They can be used for a variety of applications, whether it is professional photography, videography or for inspection (surveillance)

HOW IS THE CAMERA MOUNTED ON THE NACELLE

Before the camera is mounted on the nacelle (Gimbal), its first all place in a **flexible camera support**

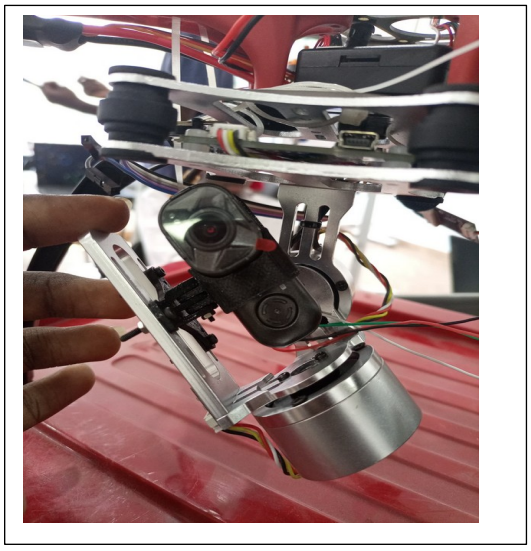




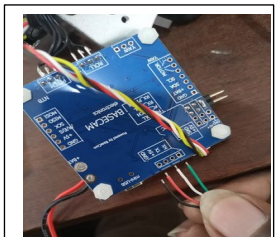
Then now using the flexible camera support, we're going to fix the camera to the nacelle in an equilibrium position and verify the function of the camera. Here it follows;





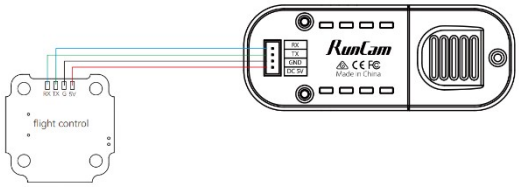


Now, **connecting the camera to the nacelle (camera support);**

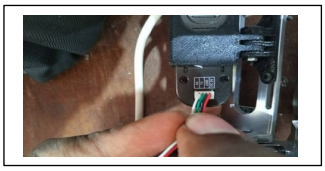


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Power supply connection and flight controller connection



Please avoid long video recording in the case of high temperature and no ventilation.
Overheating will affect the performance of the device and cause the product to operate abnormally.



Requires welding

PLACING THE NACELLE(GIMBAL) ON THE DRONE ALREADY WITH THE CAMERA

We first have to verify if the nacelle is compatible with the drone, then using attachment we can attached it firmly or screw firmly on the drone. As shown below;



Now, we move to phase 2

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PHASE 2: CALIBRATION AND AUTONOMY CALCULATIONS

In this phase, our main objective was to finalize the calibration procedure of the various sensors and ESCs. We carried out extensive tests to ensure the accuracy and reliability of the sensors used. In addition, we used mechanical and energy data to refine the calculation of battery life. This calculation allowed us to determine the electrical power needed to take off the drone to keep it in flight

I-CALIBRATION OF SENSORS AND ESCs

To allow the drone to fly correctly and according to the desired directions, we had calibrated some sensors on the drone (sensors for the direction and location of the drone in flight). These sensors are the Compass (boussole) and the accelerometer, we have also calibrated the radio control and the ESCs. To be able to do all these calibrations we used the Mission planner application.

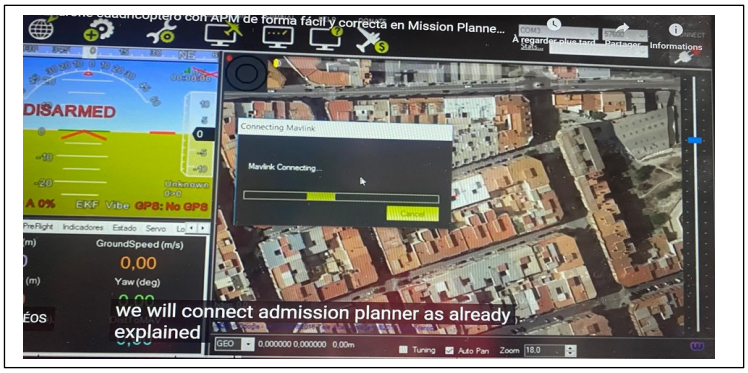
A-COMPASS CALIBRATION

Calibration of the compass (or compass) for a drone consists of providing accurate information on the magnetic orientation of the drone in relation to the Earth's magnetic field. This allows the drone to know its direction and angle of orientation when flying.

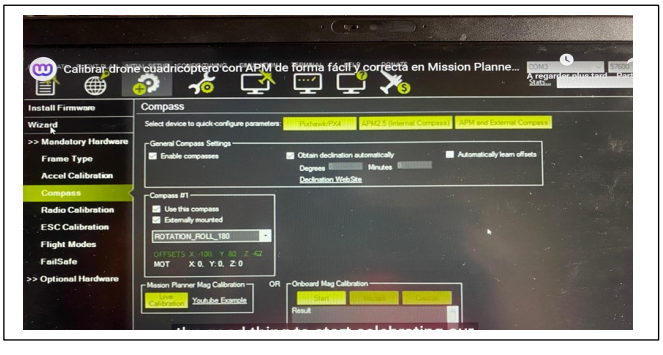
NOTE: DO NOT calibrate the compass in locations where magnetic interference may occur, such as close to magnetite deposits or large metallic structures such as parking structures, steel reinforced basements, bridges, cars, or scaffolding. Avoid calibrating near cell towers. Do NOT carry objects (such as cell phones) that contain ferromagnetic materials near the aircraft during calibration.

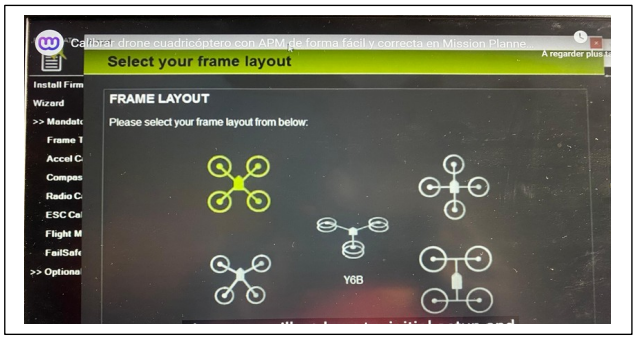
These are the steps to follow for the calibration of the compass, Choose an open area to carry out the following procedure:

1-First connect the Quadri copter to the computer and connect the battery. We will connect the admission planner and start to calibrate.



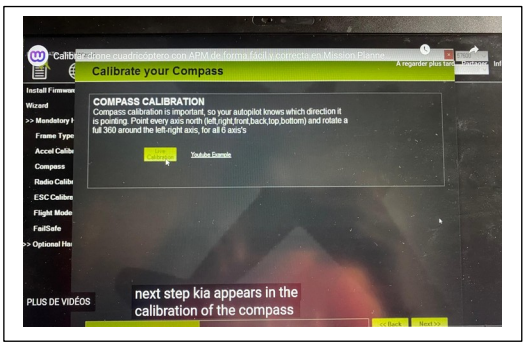
2-Then we move to initial setup and then select WIRZARD, on the screen will appear different frame and we are going to select the frame in the shape X as a Quadri copter as below;

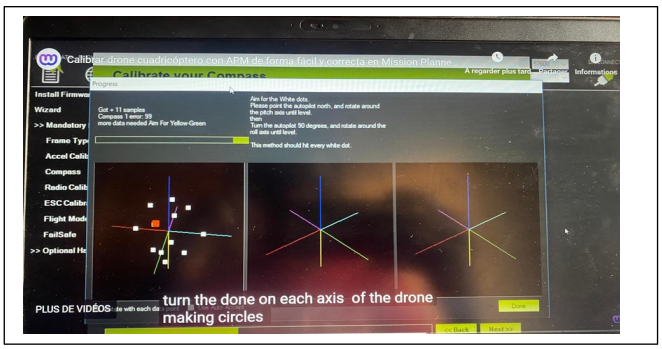




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3-Next we going to select compass calibration and select live calibration then OK, then we have to follow the instructions required. What we have to do is to turn the drone on each axis of the drone making circles in the best possible way that you can since the USB cable is involved.





4-Hold the aircraft horizontally and rotate it 360- degrees. The Aircraft Status Indicator will turn solid green.

5. Hold the aircraft vertically, with its nose pointing downward and after pointing upward, and rotate it 360 degrees around a vertical axis.

6. Then place the drone on a horizontal surface and tap “Finish calibration”. Leave the drone for a while to finish the calibration.

7. If the aircraft status indicator blinks red, the calibration has failed. Change your position and restart back the process.

8. “Finish calibration” button will appear when all the calibration movements are completed

CALIBRATION COMPLETED!

Rotating horizontally

Rotating vertically

Rotating side ways

Horizontal surface

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B- CALIBRATION OF ACCELEROMETER

Accelerometer calibration establishes a reference point for the drone's level position and ensures that the drone maintains stability during flight. Incorrect accelerometer calibration can lead to imprecise flight control responses, making it challenging to maintain a steady hover or execute precise maneuvers

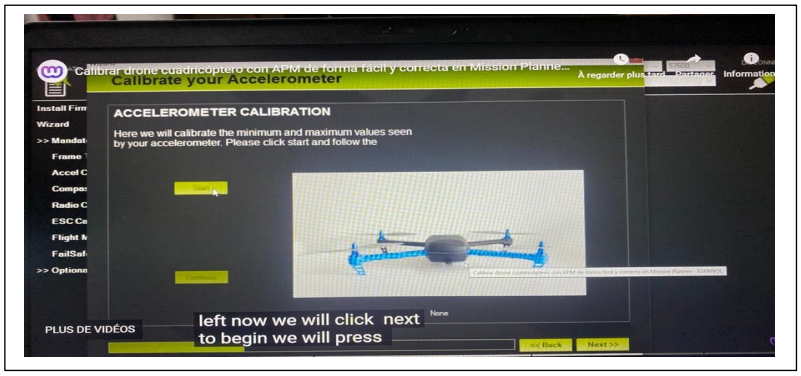
NOTE: Accelerometers must be calibrated yearly to ensure the accuracy of the data they are collecting. Calibrations are generally acceptable for a one-year term then they must be calibrated again. This process must be continued for the entire life of the sensor.

These are the steps to follow for the calibration of the accelerometer, choose an open area to carry out the following procedure;

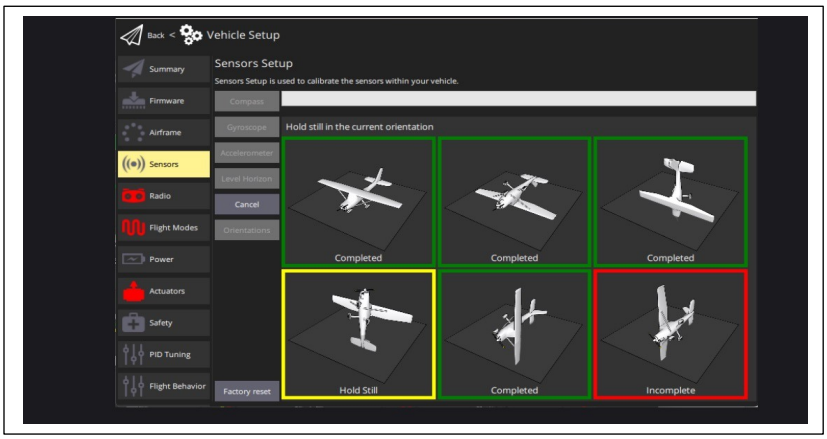
1-Step one and step two are repeated as above for the compass calibration.

2. Position the accelerometer with the arrow pointing down for the first calibration point.

3-Next we are going to select the accelerometer calibration sensor button and select START then OK, then we have to follow the instructions required.



4-When clicked on start then, Position the vehicle as guided by the images on the screen. Once prompted (the orientation-image turns yellow) hold the vehicle still. Once the calibration is complete for the current orientation the associated image on the screen will turn green. Then NEXT and follow the same procedure for different positions asked.



CALIBRATION SUCCESSFUL!

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C-RADIOLINK R9DS CALIBRATION AND CONNECTION

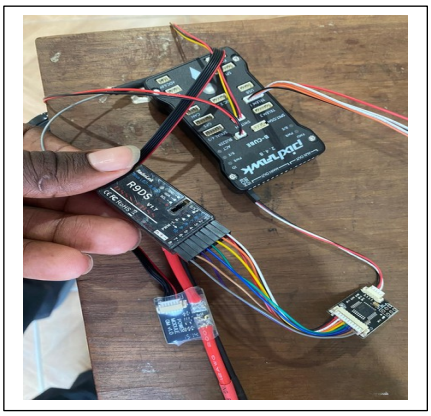
R9DS is a receiver with different channels for transmission of signals. This calibration is to connect the R9DS (receiver) to the Radio link command control (transmitter).

Pairing the signal with the transmitter:

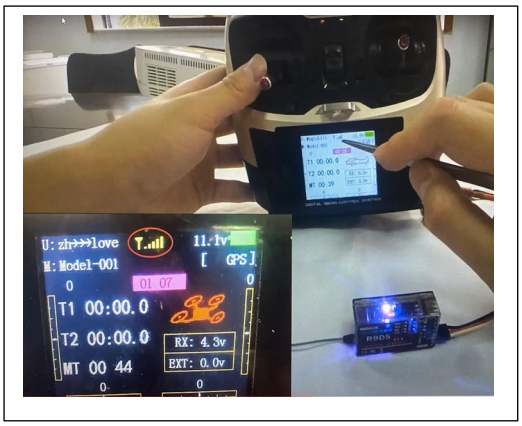
1. Place the transmitter and receiver within one meter

2. Turn on the transmitter and then the receiver

3. The receiver has a black binding button. Press it twice within two seconds. The LED will flash. After about eight flashes, the diode will start to burn with a constant light, thus signaling the correct pairing of the transmitter with the receiver.



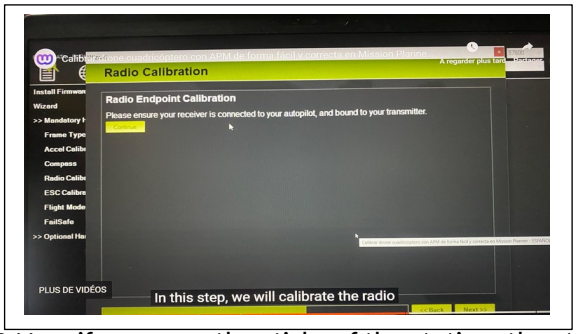




Now we're moving to the procedure of the calibration of the radio, these are the steps to follow;

1.Click on radio calibration on the screen in the setup

2.In this step we will calibrate the radio in our box, then pick up the station, we will turn it on and then enter and click on continue as you can see below.

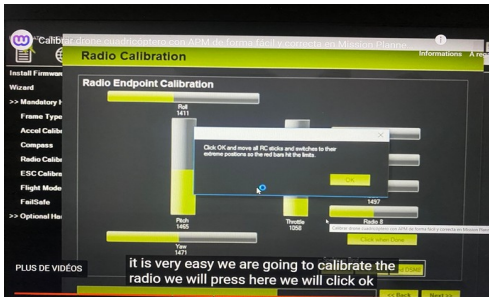


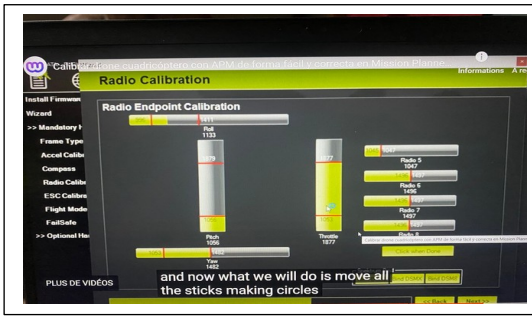
3.Now if we move the sticks of the station, the sticks broadcast planner's bars move up, down and to the sides. Before calibrating the radio, it is very important that you do not pull the right stick up and here in broadcast planner the pitch bar, when you pull the right stick up it should go down and when you pull the right stick down it should go up the pitch, if you have it like that then the radio is configured correctly. If it goes the other way round then you have to change it in the station itself.



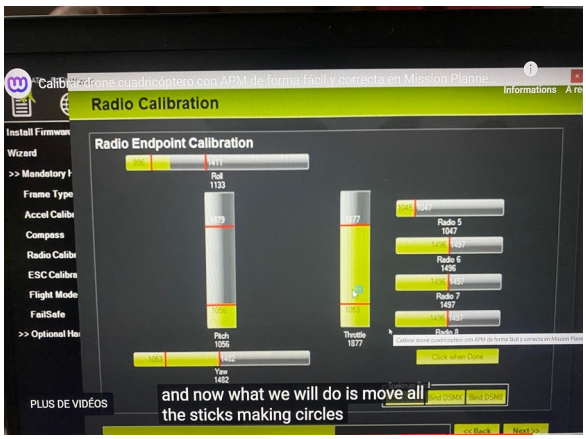
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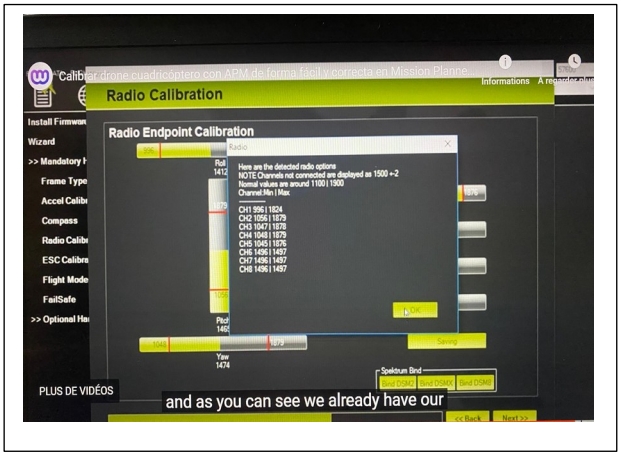
4-But then as we calibrate the radio, it is very easy we are going to calibrate the radio we will press on calibrate radio and then OK and now what we will do is to move all the sticks making circles. So that they detect the maximum positions of the station.





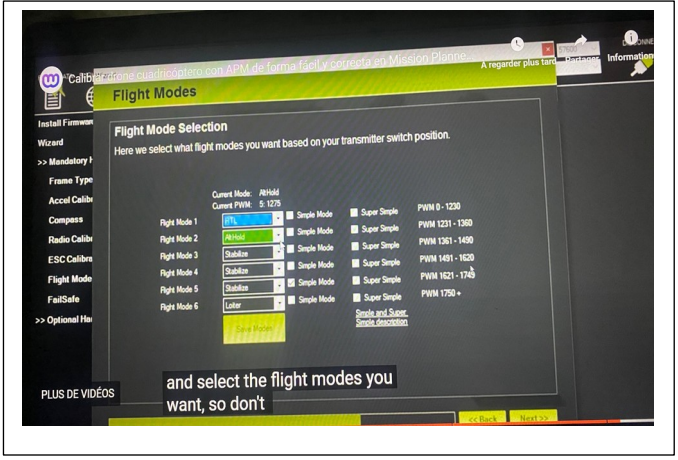
5- We will do the stick's path toward all its directions and once we have it, we will also move the flight mode selection switches so that they are also calibrated. What we have to do before clicking where we will have some sticks, the gas and direction sticks to the center so that the radio will have calibrated completely. Then going to click on “click when done” and then OK.





6-Ater the click on OK will appear the different flight mode where you have to change the flight modes on the station and select the one you want.

CALIBRATION DONE!!



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Moving lastly on ESCs calibration,

D) ESCs CALIBRATION

ESC calibration will vary based on what brand of ESC you are using, so always refer to the documentation for the brand of ESC you are using for specific information (such as tones). "**All at once**" calibration works well for most ESCs, so it is a good idea to attempt it first and if that fails try the "**Manual ESC-by-ESC**" method.

NOTE: Before calibrating ESCs, please ensure that your copter has NO PROPS on it and that the autopilot is NOT CONNECTED to your computer via USB and the Lipo battery is disconnected.





Here we are going to talk about three types of ESCs calibration which are; All at once calibration, Manual ESCs by ESCs calibration, and Semi-automatic ESCs by ESCs calibration. But the calibration that fits us was **Manual ESCs by ESCs calibration**.

Let start with the first calibration "ALL AT ONCE CALIBRATION",

A) **ALL AT ONCE CALIBRATION**

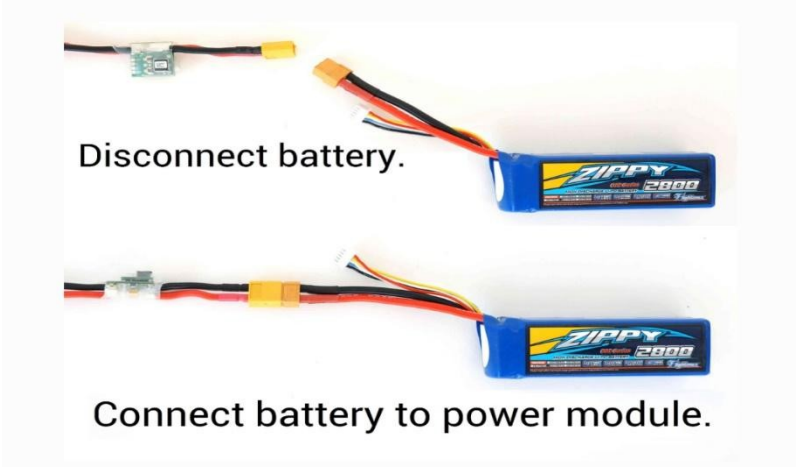
1- Turn on your transmitter and put the throttle stick at maximum.



2- Connect the Lipo battery. The autopilot's red, blue and yellow LEDs will light up in a cyclical pattern. This means the it's ready to go into ESC calibration mode the next time you plug it in.



3- With the transmitter throttle stick still high, disconnect and reconnect the battery.



4- For Autopilots with a safety switch, push it until the LED displays solid red

5- The autopilot is now in ESC calibration mode.

6- Wait for your ESCs to emit the musical tone, the regular number of beeps indicating your battery's cell count (i.e. 3 for 3S, 4 for 4S) and then an additional two beeps to indicate that the maximum throttle has been captured.

7- Pull the transmitter's throttle stick down to its minimum position.



8- The ESCs should then emit a long tone indicating that the minimum throttle has been captured and the calibration is complete.

9- If the long tone indicating successful calibration was heard, the ESCs are "live" now and if you raise the throttle a bit they should spin. Test that the motors spin by raising the throttle a bit and then lowering it again.

10- Set the throttle to minimum and disconnect the battery to exit ESC-calibration mode.

B) Manual ESC-by-ESC CALIBRATION

1- Plug one of your ESC three-wire cables into the throttle channel of the RC receiver. (This is usually channel 3.)

2- Turn on the transmitter and set throttle stick to maximum (full up).

3- Connect the LiPo battery

4- You will hear a musical tone then two beeps.

5- After the two beeps, lower the throttle stick to full down.

6- You will then hear a number of beeps (one for each battery cell you're using) and finally a single long beep indicating the end points have been set and the ESC is calibrated.

7- Disconnect battery. Repeat these steps for all ESCs.

8-If it appears that the ESC's did not calibrate then the throttle channel on the transmitter might need to be reversed.

9-If you are still having trouble after trying these methods (for example, ESCs still beep continuously) try lowering your throttle trim 50%.

C- Semi-automatic ESCs by ESCs calibration

1-Connect to the autopilot from a ground station such as the Mission Planner and set the ESC_CALIBRATION parameter to 3

2- Disconnect the battery and USB cable so the autopilot powers down

3- Connect the battery.

4- The arming tone will be played (if the vehicle has a buzzer attached)

5- If using an autopilot with a safety button (like the Pixhawk) press it until it displays solid red

6- You will hear a musical tone then two beeps.

7- A few seconds later you should hear a number of beeps (one for each battery cell you're using) and finally a single long beep indicating the end points have been set and the ESC is calibrated

8- Disconnect the battery and power up again normally and test as described below

TESTING

Once you have calibrated your ESCs, you can test them by plugging in your LiPo. Remember: no propellers!

-Ensure your transmitter's flight mode switch is set to "Stabilize Mode".

-Arm your copter

- Give a small amount of throttle. All motors should spin at about same speed and they should start at the same time. If the motors do not all start at the same time and spin at the same speed, the ESCs are still not properly calibrated.

-Disarm the drone

CALCULATING THE AUTONOMIE OF THE DRONE IN FLIGHT

This calculation allowed us to determine how long the battery will last when the drone is in flight and the energy needed for the flight. The main source of energy used to power the drone is the **battery.**

NOTE: The battery should be charged completely and should not be discharged completely before recharging. The lifespan of a battery is approximately 300 charges.

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CALCULATIONS WITH BATTERY

Charge of battery(Q)= 6000mAh

Voltage (Tension (V)) =11.5v

BEC (Battery Eliminator Circuit) of an ESC=3A

Calculating the power consumption to lift

For one motor

P=IV, P=11.5(3)

=34.5W

For all motors

P=34.5*4

=138W

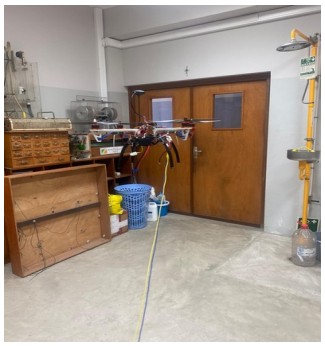
AUTONOMIE= Charge(Q)/I= 6/6 = 1h

So normally our battery allows a current of 6000mA for one hour

CALCULATIONS WITH GENERATORS

We connect 3 generators in series and input the voltage at 11.1V. Then connect the drone and make it flight to register the intensity of current during the flight for the calculation.





So, during the flight, V=12.5v, I=6.36+10.2+5.55

=22.1A

AUTONOMY of drone= Q/I

=6Ah/22.1A

=0.27h, Converting in minutes

= 0.27*60= 16min

But calculating with the new battery that we made,

Q=4400mAh, I=22.1

New AUTONOMY= Q/I

=4.4Ah/22.1A

=0.199h

=0.199*60

=12min

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